MIL-B-36170A
24 March 1970
SUPERSEDING
MIL-B-36170(DM)
12 November 1963

MILITARY SPECIFICATION

BRUSH, POLISHING, DENTAL HANDPIECE (RIGHT ANGLE)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers screw stem type polishing brushes suitable for use with right angle prophylaxis handpieces in cleaning and polishing teeth.

2. APPLICABLE DOCUMENTS

2.1 Specifications, standards, and drawing. The following documents of the issue in effect on date of invitation for bid or requests for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water-Resistant (For Closure of Fiberboard Boxes).
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 Marking for Shipment and Storage.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

20676 Brush, Polishing, Dental Handpiece

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

* 3.1.1 Brush stock. The brush stock shall be top grade, stiff, natural, undyed, black horsehair. The hair shall be straightened, thoroughly mixed, and cleaned. The ends of the hair shall be square cut.

3.1.2 Metal components. The hub and screw stem shall be fabricated of free-machining brass. The hub section shall be nickel plated. Nickel plating of the screw stem is optional. When staples are used for clinching the hair, the staples shall be made of corrosion-resisting metal, or metal suitably coated to render it corrosion-resisting.

* 3.2 Style, design and dimensions. The brushes shall be of the style, design, and dimensions shown for style 1, 2, or 3 brush of Defense Personnel Support Center (DPSC) Drawing 20676 and shall be

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suitable for use as prophylaxis polishing brushes. The polishing brushes shall be suitable for use with short right angle prophylaxis handpieces conforming to the requirements of Federal Stock Number (FSN) 6520-935-7141. The hair shall be securely and permanently clinched and fastened in the hub by means of a staple around the middle of the bent tuft. As an alternate, straight hair may be securely glued into the hub. The hair shall be tightly packed in the hub and neatly and evenly trimmed. The brush shall retain the hair, when tested as specified in 4.4.1 and 4.4.2.

3.3 Corrosion-resistance. The metal components shall show no corrosion when tested as specified in 4.4.2.

3.4 Finish. The exposed metal components of the brush shall have a bright, smooth finish and shall be free from rough edges which may be injurious to gum tissue.

3.4 Workmanship. Workmanship shall be first class throughout. Brushes shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

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* 4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the horsehair and brass, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the name of the material, the lot number, the test method, individual test results, and the specification requirements.

* 4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in Tables I and III (see 4.2.1 and 4.3) and Table II (see 4.2.2).

* 4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. Unit of product for examination purposes shall be 6 brushes from each unit of issue (24 brushes).

Table I. Inspection and Acceptable Quality Levels (AQL's)

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.0
Major B	S-2	2.5
Minor	II	4.0

* 4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II, except that the acceptable number shall be zero for all sample sizes. Unit of product for test purposes shall be 6 brushes from each unit of issue (24 brushes).

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Table II. Tests

Characteristic	Requirement	Test Procedure	Inspection level
Hair retention	3.2	4.4.1 & 4.4.2	S-1
Corrosion	3.3	4.4.2	S-1

* 4.3 Examination. Brushes shall be examined for defects, including but not limited to, those listed in Table III.

Table III. Classification of Defects

Categories	Defects
Major A:	
101	Style and design not as specified.
102	Color of hair not as specified.
103	Hair not stiff, natural, and undyed.
104	Hair not straight.
105	Ends of hair not square cut and evenly trimmed.
106	Hair not tightly packed in hub.
107	Hair not secured in hub by means of a staple or glue.
108	Rough edges present on exposed metal surfaces of brush.
Major B:	
151	Dimensions not as specified for specific style.
152	Count in unit package not as specified.

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Table III. Classification of Defects (cont'd.)

Categories	Defects
Minor:	
201	Finish of metal surfaces not bright.
202	Packaging and packing not as specified.
203	Marking not as specified.

4.4 Tests.

4.4.1 Hair retention. The brush shall be rotated at 2,500 revolutions per minute and applied to a metal disc under a 300-gram load for 5 minutes. The brushes shall be examined for hairs that may be loosened.

4.4.2 Boil test. The brushes shall be scrubbed with soap and water and rinsed thoroughly in distilled water. After cleaning, the brushes shall be boiled in distilled water in a glass container for 30 minutes and allowed to remain submerged for 24 hours. The specimens shall then be removed from the water and examined for corrosion and retention of the hair.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package (24 brushes), as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	864 units

5.1.2.1 Variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable. Closure shall be adequate to prevent accidental opening under normal handling.

* 5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2-inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted.

* 5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing.

5.2.1 Level A.

* 5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

* 5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

* 5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

* 5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Federal Stock Number (FSN) covered. This specification covers the requirements for FSN 6520-890-1565, BRUSH, POLISHING, DENTAL HANDPIECE, Natural Bristle, 24s: Screw Stem.

6.3 Specification changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.4 Limitations. This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION

MIL-B-36170A FSN:

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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